# Alejandro Vasquez Jimenez

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Advanced Electronics Engineering student at Instituto Tecnologico de Costa Rica, expected graduation 2026, focused on validation workflows, applied software tools, data analysis, and practical ML systems. Project experience includes a modular computer vision pipeline, AI-assisted workflow design, a data dashboard for a real stakeholder, and a digital control project on FPGA. Organized, proactive, and curious; strongest when turning technical uncertainty into structured, testable work.

## Experience

**Academic Tutor - Signals and Systems**
Instituto Tecnologico de Costa Rica | 2023, one semester

- Tutored undergraduate Electronics Engineering students through weekly reviews and one-on-one support, adapting explanations of Fourier analysis, LTI systems, and signal transforms to different skill levels.

**Electronics Department Intern**
Hultec | 2019, 3 months

- Performed component-level diagnostics and verification on industrial sensors and relays using multimeters and oscilloscopes; updated legacy electrical schematics into AutoCAD and supported preventive maintenance on machine connection boxes.

## Selected Projects

**BloodyDiagnose - Multi-stage ML Pipeline for Blood-Smear Image Analysis**
Team project, 2 members

- Built part of a modular computer vision pipeline for blood-smear images, covering detection, classification, evaluation, and report-oriented aggregation.
- Structured the project around explicit pipeline stages, artifact tracking, automated tests, and quality checks so reviewers can inspect behavior beyond a notebook-only model.

**QuantSpec - AI-Assisted Hypothesis Validation Workflow**
Self-initiated project

- Designed a spec-first workflow for turning trading hypotheses into structured briefs, validation gates, backtest artifacts, and decision-oriented reports with traceable human review.

**Vitali Pricing - Data Dashboard for a Local Host**
Self-initiated project

- Built an interactive dashboard over real reservation income data to surface demand patterns, pricing references, and monthly balance estimates for a non-technical stakeholder.

**FPGA Digital Control System for a DC-DC Boost Converter**
Team project, 3 members

- Contributed to the design and validation of a digital control system for a power-electronics application, using a modular SystemVerilog implementation on Basys3.
- Helped structure verification before hardware bring-up through simulation, test procedures, peer review, and traceable documentation.

## Education

**Instituto Tecnologico de Costa Rica (ITCR)**
Bachelor's in Electronics Engineering | Cartago, Costa Rica | Expected graduation: 2026
Relevant coursework: Machine Learning and Data Science.

**COVAO**
Technical Diploma in Industrial Electronics | Cartago, Costa Rica | Graduated: 2019

## Technical Skills

**Core:** Python, MATLAB / Octave, Git, Linux, LaTeX.
**Validation and systems:** test planning, simulation-based verification, traceable documentation, hardware-oriented debugging.
**Data and ML:** data analysis workflows, computer vision pipelines, dashboards, automated tests, reproducible project structure.
**Electronics foundation:** SystemVerilog, FPGA design, lab instrumentation, industrial electronics.
**Familiar:** C, C++, AutoCAD.
**Languages:** Spanish native, English advanced, Portuguese basic.